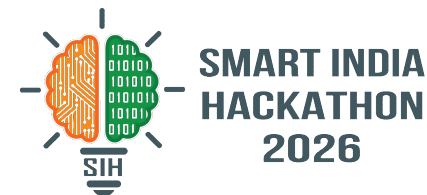




SMART INDIA HACKATHON 2026



Problem Statement ID –*SIH26117*

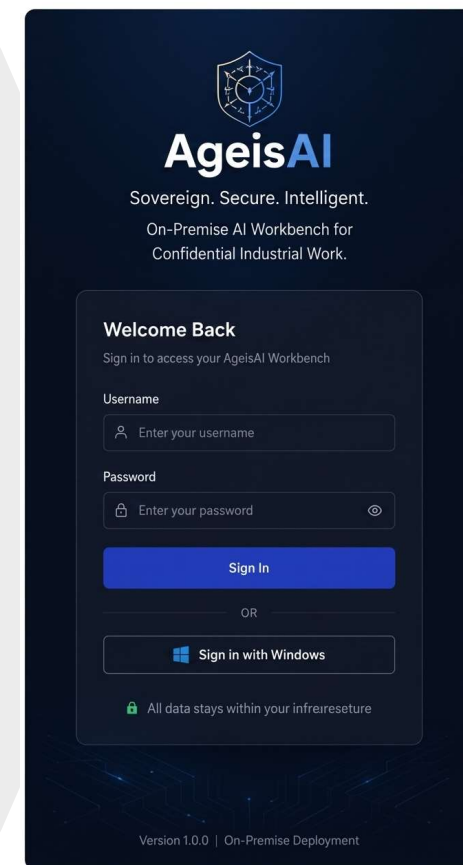
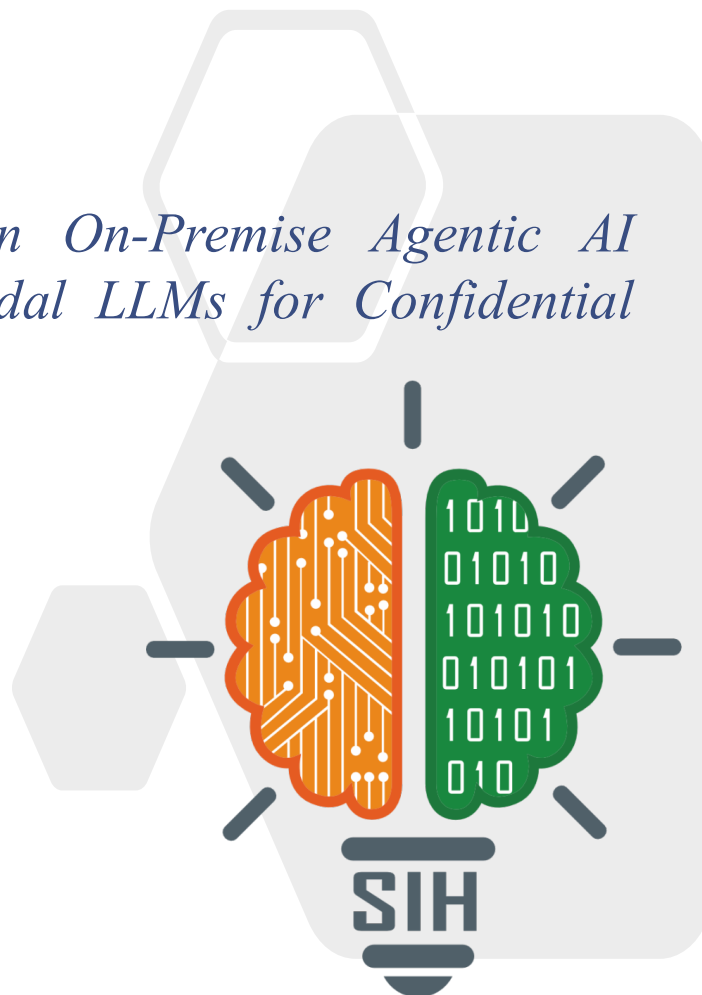
Problem Statement Title- *Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work.*

Theme- *Smart Automation*

PS Category- *Software*

Team ID-

Team Name:*GLITCH*



1. LOGIN SCREEN

Secure accers to AgeisAI Workbench



AegisAI

Sovereign Agentic AI for Secure Industrial Workflows



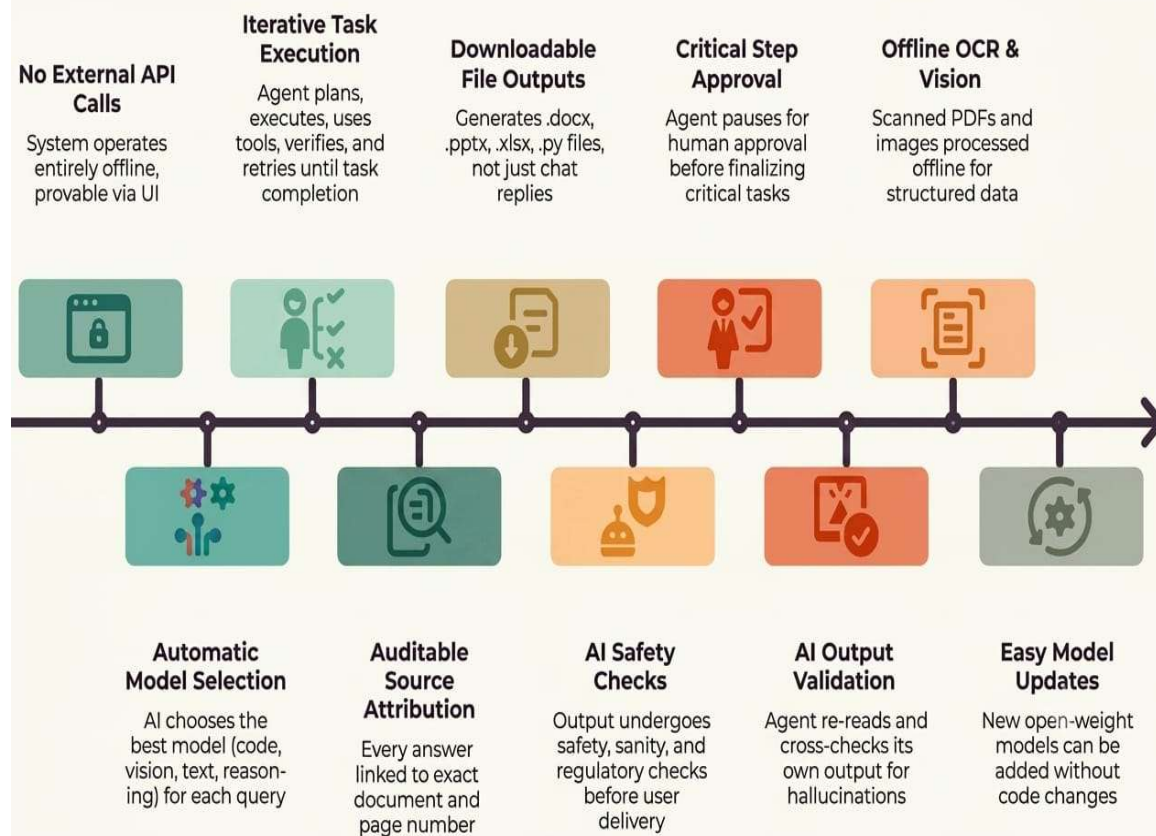
PROBLEM OVERVIEW

- **Confidential industrial/government data** cannot be sent to cloud AI tools.
- Employees face two choices: **manual work or unauthorized AI use.**
- **Current on- premise AI** = single- model chatbots, limited capability.
- **They fail at multi- step tasks:** document understanding, reasoning, tool use.
- **Need:** secure on- premise AI workbench.
- Must handle **full industrial knowledge work** inside organization.
- **Key features:** data sovereignty, multi- model orchestration, workflow automation, compliance.

PROBLEM SOLUTION

- **Ageis AI:** Runs entirely on-premise with zero external API dependency.
- **Multimodal Intelligence:** Processes industrial documents, images, drawings and technical data.
- **Trusted RAG:** Generates citation-backed answers using verified internal knowledge.
- **Agentic Automation:** Executes multi-step industrial tasks through controlled AI agents.
- **Intelligent Routing:** Automatically selects the most suitable local model for each task.
- **AI Safety Layer:** Combines risk assessment, self-verification, sandboxing and human approval

INNOVATION AND UNIQUENESS



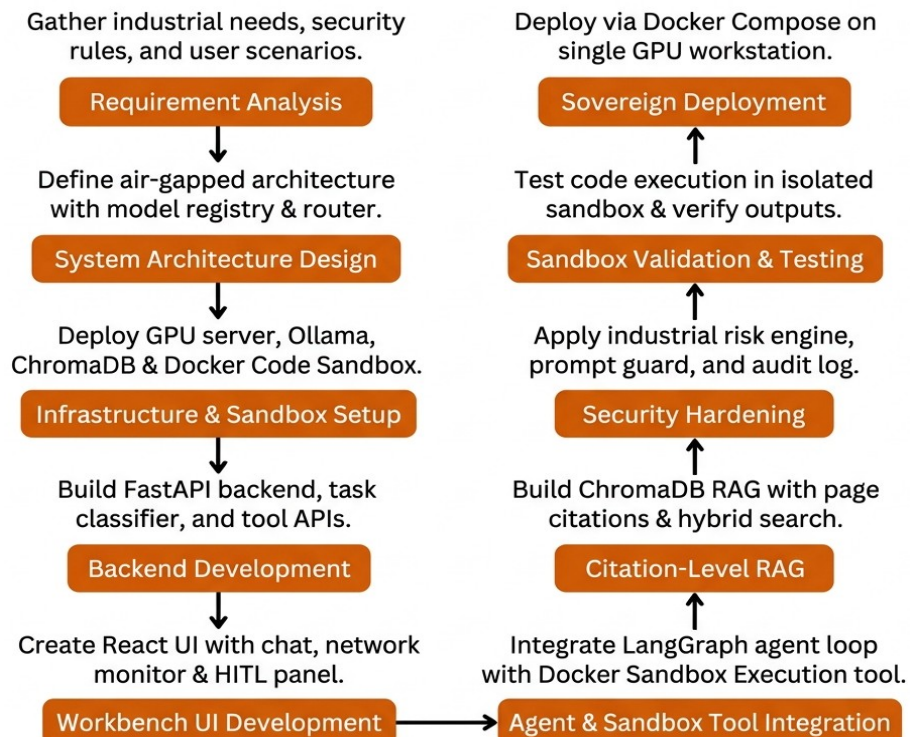


TECHNICAL APPROACH

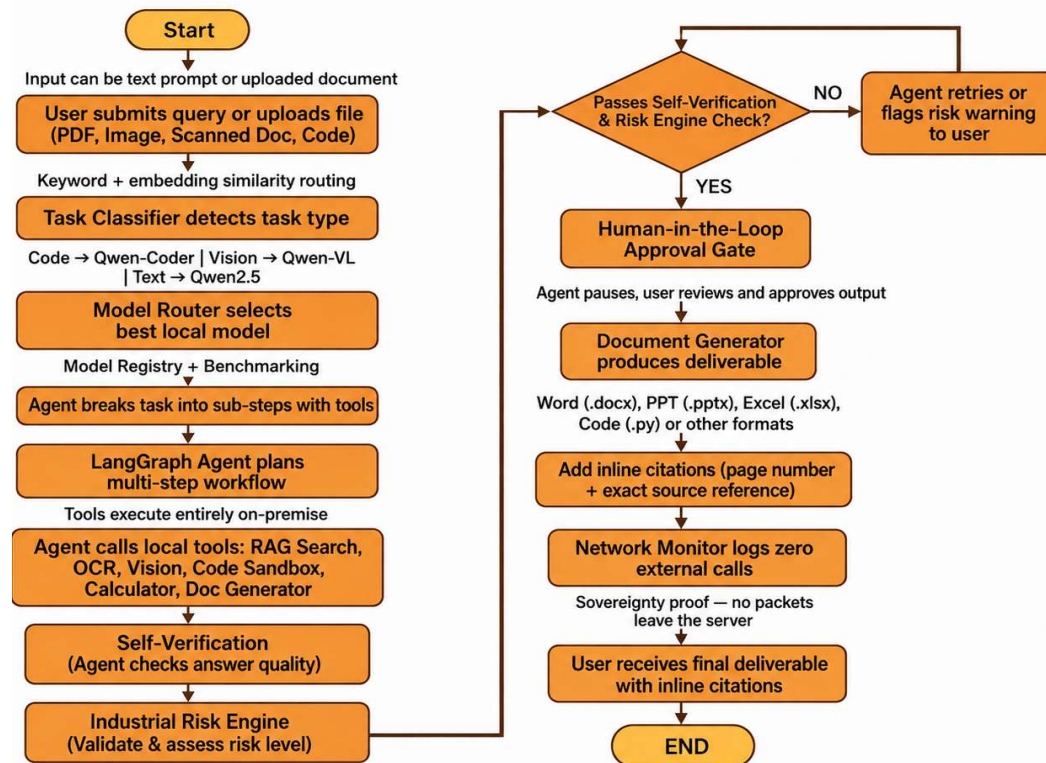


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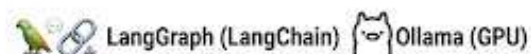
METHODOLOGY



FLOW CHART:

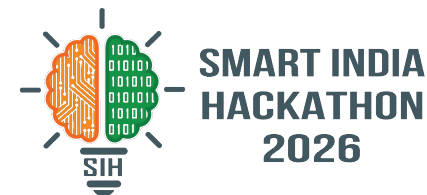


OUR COMPLETE AGENTIC TECH STACK





FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA

1. TECHNICAL FEASIBILITY

Achievable on a mid-range GPU workstation using Ollama model runner, LangGraph agent framework, FastAPI backend, and ChromaDB vector RAG.

2. ECONOMICAL FEASIBILITY

Zero recurring API costs (unlike Claude/Codex). Low total cost of ownership by running open-weight 7B models locally on existing PSU hardware.

3. SOCIAL FEASIBILITY

Boosts productivity for PSU and defense engineers while ensuring 100% data confidentiality, preventing unauthorized use of public AI tools.

4. LEGAL & COMPLIANCE FEASIBILITY

Architecture enforces local data processing with no external API dependency and strict compliance with PSU/defense air-gap policies, IT Act, and CERT-In security guidelines.

5. OPERATIONAL FEASIBILITY

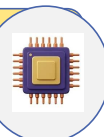
Simple web interface, automated model selection, and human-in-the-loop approval ensure seamless adoption without needing deep AI expertise.

6. SECURITY FEASIBILITY

Isolated Docker sandbox, local RBAC, on-device OCR, and real-time Network egress controls and monitoring are designed to prevent unauthorized external communication

POTENTIAL CHALLENGES AND RISKS

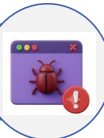
Running multimodal and coding models on limited VRAM hardware



Extracting text from dense P&ID drawings, handwritten notes, and scanned PDFs.



Malicious inputs attempting to bypass safety guardrails or access unauthorized documents.



Risk of AI generating incorrect engineering figures or formulas.



STRATEGIES FOR OVERCOMING THESE CHALLENGES

MODEL AUTO-ROUTING



Auto-route tasks to single 7B quantized models via Ollama to run efficiently on 12GB VRAM.

HYBRID OCR & VISION



Surya OCR combined with Qwen2-VL for accurate text and engineering diagram parsing.

INDUSTRIAL RISK ENGINE



Agent self-verification & rule-based sanity checks to validate calculations before delivery.

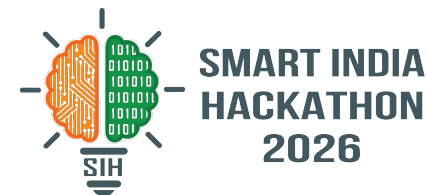
STRICT GUARDRAIL SANITIZER



Regex input filtering, RBAC document tags, network activity monitoring and real-time logging.



IMPACT AND BENEFITS



UNIQUE FEATURES:

Provable Air-Gapped Sovereignty

Real-time network monitor proves zero external packet transmission.

Multi-Model Task Auto-Routing

Hybrid classifier automatically selects the optimal open-weight model per task.

Plan-Execute-Verify Agentic Loop

Autonomous multi-step execution using Docker Sandbox, RAG, and OCR tools.

Citation-Level RAG Grounding

Every response is grounded with exact document and page-level source attributions.

Industrial Risk Engine & HITL

Output safety validation and mandatory human-in-the-loop approval gates.

IMPACTS

Zero-Trust Air-Gapped OPSEC



Confidential industrial data stays on-premise, with network monitoring verifying zero external communication.

Knowledge Workflow Acceleration



Automates document parsing, SOP cross-referencing, calculations and report generation to drastically reduce manual effort.

Citation-Level Grounding



Every answer and recommendation is linked to the exact source document and page for verification.

Multimodal Industrial Perception



Processes scanned PDFs, handwritten logs, equipment images and engineering drawings completely offline.

BENEFITS

Eliminate Cloud AI Costs



Runs open-weight models locally, reducing recurring cloud API costs and vendor dependency.

Security & Compliance



Combines air-gapped deployment, access controls, encryption and audit trails for sensitive industrial environments.

Future-Proof Model Portability



Offline model registry and intelligent routing allow new open-weight models to be added easily.

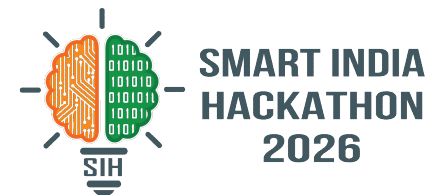
Turnkey Deployment



Docker-based deployment simplifies installation while isolated sandboxes safely execute agent tools.



RESEARCH AND REFERENCES



The screenshot displays the AgeisAI AI Workbench interface. On the left is a sidebar with navigation options like Dashboard, Knowledge Base, Documents, Models, Agents, and Tools. The main area shows a conversation history with a recent query: "Please analyze this P&ID and let's oil safety relief valves with their set pressure and discharge the details." The analysis results are shown in a table:

SRV ID	Set Pressure	Discharge Line	Connected To	Location
SRV-101	10.5 barg	3" Line to Flare Header	Boiler Drum	North Side
SRV-102	9.8 barg	2" Line to Vent	Steam Header	East Side
SRV-103	11.0 barg	3" Line to Flare Header	Superheater	West Side

Below the table, it states: "All relief valves are in compliance with ASME Section VIII standards." To the right, a document viewer shows a P&ID drawing with highlighted results.

2. WORKBENCH — ANALYZE & INTERACT
Chat with documents, get analysis with citations

3. DOCUMENT VIEWER - CITATIONS
View source document with highlighted results

This screenshot shows the AgeisAI Sovereign On-Premise Agentic AI Workbench interface. It includes a sidebar with navigation options and a main area with a conversation history, analysis results, and a document viewer. The interface also features a system status section with metrics like CPU, GPU, RAM, and Storage. A model router section shows a list of models and their status. An agent output section displays the results of the analysis.

Interface

- **Qwen2.5-Coder & Qwen2-VL Models**
Summary: Open-weight 7B foundation models delivering SOTA code generation and multimodal parsing of scanned PDFs and P&ID engineering diagrams.
Link: <https://huggingface.co/Qwen/Qwen2.5-Coder-7B-Instruct>
- **ReAct & Self-Refine Agent Architecture**
Summary: Framework combining step-by-step reasoning with tool execution, featuring autonomous self-verification loops before output delivery.
Link: <https://arxiv.org/abs/2303.17651>
- **LangGraph & ChromaDB RAG Framework**
Summary: Stateful graph orchestrator enabling Human-in-the-Loop review paired with a zero-dependency, local vector store for page-level PDF citations.
Link: <https://www.sciencedirect.com/science/article/pii/S0957417426000746>
- **Hybrid BM25 & Dense Vector Retrieval**
Summary: Combines keyword search with semantic embeddings (Sentence-BERT) to guarantee exact document citation matching for technical SOPs.
Link: <https://doi.org/10.1145/3471158.3472233>
- **CERT-In & NCIIPC Cybersecurity Guidance**
Summary: Government cyber security standards mandating 100% on-premise data containment and zero outbound network telemetry for Critical Information Infrastructure.
Link: <https://www.cert-in.org.in/s2cMainServlet?pageid=GUIDLNVIEW02&refcode=CISG-2023-01>